

Task -2:

Objective:

The goal of this task is to enhance the **UI/UX** of the previously completed functional prototype, while also making improvements to the **backend logic** and **database structure**. The end result should be a **beautiful, intuitive, and responsive frontend** that integrates seamlessly with the backend.

1. UI/UX Enhancements

1.1 Authentication Flow (Login/Registration):

- Implement smooth and seamless **login** and **registration** pages with proper validation and feedback.
- Use **JWT-based authentication** and ensure role-based redirection after login.
- Provide visual feedback like form validation, loader animations, and error handling during authentication.

1.2 Dashboard (Student & Professor Views):

- Design the **student dashboard** to display all enrolled courses.
- Each course should be clickable to open the respective assignment page.
- Ensure responsive layout using grid systems for course cards.
- Design the **professor dashboard** to show courses being taught along with basic analytics such as submission statuses and student count.

1.3 Assignment Page:

- Display assignment details such as title, description, deadline, and submission type (Individual/Group).
- Show clear prompts and visual feedback for submission status, including progress bars, acknowledgment status, and animation for changes.
- For **group assignments**, implement logic where only the group leader can acknowledge submission, which should be reflected across all group members.

1.4 Progress Visualization:

- Use **progress bars** and **badges** to display submission progress.
- Implement visual feedback like **checkmarks** and **subtle animations** on submission completion.
- Ensure the UI communicates submission status clearly and intuitively for both individual and group submissions.

2. Database Enhancements

2.1 Database Structuring:

- Ensure the database schema is properly structured to support the needs of both students and professors.
- Design the **students** and **professors** collections to include relevant fields like enrolled courses, assignments, and submission status.
- Implement a **courses** collection that associates students and assignments with the correct course.
- Design the **assignments** collection to store assignment details and submission statuses for each student.
- Create a **groups** collection to handle group assignments, with references to the students and their leader.

2.2 Data Relationships:

- Ensure proper relationships between students, assignments, and courses through references.
- Implement logic to manage group memberships and their impact on assignment submissions.
- Make sure that acknowledgments are properly tracked for individual and group submissions.

3. Backend Logic

3.1 Authentication:

- Implement JWT-based **authentication** for both **students** and **professors**.
- Ensure the backend sends the correct response and redirection based on the user role (student/professor).

3.2 Assignment Management (Professor Flow):

- Enable the **professor** to **create, edit, and view assignments**.
- Allow the professor to monitor submissions, see progress, and track which students have submitted or acknowledged the assignment.
- Implement assignment status management, including the ability to filter submissions by status.

3.3 Student Assignment Flow:

- Allow **students** to submit assignments and acknowledge submissions where applicable.
- Ensure proper handling of **individual** and **group** assignment submissions, with proper validation.

- When a group leader acknowledges submission, reflect the acknowledgment across all group members.

4. Deliverables

4.1 GitHub Repository:

- Submit the **GitHub repository** with a **clear commit history** reflecting progress and changes.
- Ensure that both the **frontend** and **backend** code are in separate folders with proper organization.
- Include detailed **commit messages** to track your progress throughout the development.

4.2 Working Demo:

- Provide a **working demo** of the frontend, deployed to a hosting platform (e.g., **Netlify** or **Vercel**).
- Ensure the frontend is connected to the backend (or mock API if necessary).

4.3 README.md Documentation:

- Provide a **README.md** with the following details:
 - Overview of the UI/UX design choices and the reason behind them.
 - Setup instructions to run both the frontend and backend locally.
 - Screenshots or **GIFs** demonstrating the UI flow (login, dashboard, assignment page, etc.).
 - Optional: Include a **component structure diagram** or a description of your component architecture.

4.4 Video Demonstration (Optional but Recommended):

- Create a **detailed video** explaining the entire system, including:
 - The **UI/UX flow**.
 - How the **backend** integrates with the frontend.
 - The **database structure** and how data is stored and retrieved.
 - A **step-by-step walkthrough** of the working system.

5. Key Requirements

- **Frontend:** Use **React.js** and **Tailwind CSS** for building the UI. Ensure that the UI is clean, intuitive, and responsive across all devices.
- **Backend:** Use **Node.js**, **Express.js**, and **PostgreSQL** for the backend. Implement JWT authentication, CRUD operations for assignments, and group management logic.

- **Database:** Ensure that the database schema supports role-based workflows (students and professors) and allows efficient tracking of assignments, submissions, and group activities.
- **Deployment:** Ensure the frontend is deployed on a platform like **Netlify** or **Vercel**, and connected to a backend (either real or mock API).

SUBMISSION INSTRUCTIONS:

- **Deadline:** 29th August 2026, Saturday - 11.30 AM
- **Submission Format:** A PDF file containing:
 - GitHub repository link
 - Working demo video link
 - Deployed Platform Link(optional)
- **Filename Convention:** Your file should be named as follows: **FullName-Round2**

Example: If your name is Siddhartha Thonti, the filename should be **SiddharthaThonti-Round2.pdf**.

- **Submission Link:** <https://forms.gle/4pnM5wh1KuSrzKKk6>

We will evaluate your submission based on the work you've done. Don't give up, and submit whatever you have accomplished so far.

Next Steps

If your submission meets the expectations, you will be invited to the next Round, which will be a short live interview to walk through your work and discuss the technical aspects.

Important Notes

We value commitment and responsibility. Please ensure that:

- The task is completed and submitted within the stipulated time.
- Your solution is original, well-documented, and easy to understand.
- You are prepared for a follow-up discussion if you are shortlisted.